# PragTutor AI Platform Evaluation

## Operating Systems Multimodal AI Tutor — Real System Output Audit

This evaluation report presents **real, empirical outputs** generated directly by the running PragTutor system. The benchmark evaluates input routing (Text-Only, Text+Image, Image+OCR, Image-Only Vision), prerequisite-based personalization (Beginner, Intermediate, Advanced), RAG vector retrieval, and topic-image relevance validation.

| | |
|---|---|
| **Target Domain:** | Operating Systems (Course Corpus & Prerequisites) |
| **LLM Provider Engine:** | Active Provider via llm_service.py (.env configured) |
| **Vector RAG Store:** | ChromaDB Collection (os_knowledge_base) |
| **Image Selection Priority:** | Topic Match > Chunk Match > Page Match > Blank/White Filtered |
| **Text-Only Response Time:** | **83.794 seconds** |
| **Image-Only Vision Time:** | **14.779 seconds** |
| **Evaluation Date:** | October 04, 2026 |

## System Execution Matrix (Top 5 OS Questions across 3 Levels)

| # | Topic | Level | Prereqs | RAG Chunks | Img Valid | Topic Match |
|---|---|---|---|---|---|---|
| 1 | CPU Scheduling Algorit | Beginner | computer_system_and_software, operating_system, process, process_states | 5 | **PASS** | **PASS** |
| 1 | CPU Scheduling Algorit | Intermediate | computer_system_and_software, operating_system, process, process_states | 5 | **PASS** | **PASS** |
| 1 | CPU Scheduling Algorit | Advanced | computer_system_and_software, operating_system, process, process_states | 5 | **PASS** | **PASS** |
| 2 | Process Management & P | Beginner | computer_system_and_software, operating_system, process, process_states | 5 | **PASS** | **PASS** |
| 2 | Process Management & P | Intermediate | computer_system_and_software, operating_system, process, process_states | 5 | **PASS** | **PASS** |
| 2 | Process Management & P | Advanced | computer_system_and_software, operating_system, process, process_states | 5 | **PASS** | **PASS** |
| 3 | Process Synchronizatio | Beginner | computer_system_and_software, operating_system, process, interprocess_communication, process_synchronization | 5 | **PASS** | **PASS** |
| 3 | Process Synchronizatio | Intermediate | computer_system_and_software, operating_system, process, interprocess_communication, process_synchronization | 5 | **PASS** | **PASS** |

| | | | | | | |
|---|---|---|---|---|---|---|
| 3 | Process Synchronizatio | Advanced | computer_system_and_software, operating_system, process, interprocess_communication, process_synchronization | 5 | **PASS** | **PASS** |
| 4 | Deadlocks & Banker's A | Beginner | computer_system_and_software, operating_system, process, resource_allocation, deadlock, deadlock_characterization, deadlock_avoidance, safe_state, safe_sequence | 5 | **PASS** | **N/A (No image)** |
| 4 | Deadlocks & Banker's A | Intermediate | computer_system_and_software, operating_system, process, resource_allocation, deadlock, deadlock_characterization, deadlock_avoidance, safe_state, safe_sequence | 5 | **PASS** | **N/A (No image)** |
| 4 | Deadlocks & Banker's A | Advanced | computer_system_and_software, operating_system, process, resource_allocation, deadlock, deadlock_characterization, deadlock_avoidance, safe_state, safe_sequence | 5 | **PASS** | **N/A (No image)** |
| 5 | Memory Management & Pa | Beginner | computer_system_and_software, operating_system, process, main_memory, memory_protection, base_and_limit_registers, address_binding, contiguous_memory_allocation, variable_partition, external_fragmentation, paging, page_and_frame, page_table, address_translation | 5 | **PASS** | **N/A (No image)** |

| | | | | | | |
|---|---|---|---|---|---|---|
| 5 | Memory Management & Pa | Intermediate | computer_system_and_software, operating_system, process, main_memory, memory_protection, base_and_limit_registers, address_binding, contiguous_memory_allocation, variable_partition, external_fragmentation, paging, page_and_frame, page_table, address_translation | 5 | **PASS** | **N/A (No image)** |
| 5 | Memory Management & Pa | Advanced | computer_system_and_software, operating_system, process, main_memory, memory_protection, base_and_limit_registers, address_binding, contiguous_memory_allocation, variable_partition, external_fragmentation, paging, page_and_frame, page_table, address_translation | 5 | **PASS** | **N/A (No image)** |

# Question 1: CPU Scheduling Algorithms

**Query:** *"What is CPU Scheduling, and how do FCFS, SJF, and Round Robin algorithms optimize execution time?"*

### Level: BEGINNER (Response Time: 4.67s)

**Identified Topic:** cpu_scheduling | **Prerequisites:** computer_system_and_software, operating_system, process, process_states

**RAG Chunks Retrieved:** 5 chunks

[Gemini LLM Error HTTP 404]: {
"error": {
"code": 404,
"message": "models/gemini-1.5-flash is not found for API version v1beta, or is not supported for generateContent. Call ModelService.ListModels to see the list of available models and their supported methods.",
"status": "NOT_FOUND"
}
}

### Level: INTERMEDIATE (Response Time: 4.359s)

**Identified Topic:** cpu_scheduling | **Prerequisites:** computer_system_and_software, operating_system, process, process_states

**RAG Chunks Retrieved:** 5 chunks

[Gemini LLM Error HTTP 404]: {
"error": {
"code": 404,
"message": "models/gemini-1.5-flash is not found for API version v1beta, or is not supported for generateContent. Call ModelService.ListModels to see the list of available models and their supported methods.",
"status": "NOT_FOUND"
}
}

### Level: ADVANCED (Response Time: 4.841s)

**Identified Topic:** cpu_scheduling | **Prerequisites:** computer_system_and_software, operating_system, process, process_states

**RAG Chunks Retrieved:** 5 chunks

[Gemini LLM Error HTTP 404]: {
"error": {
"code": 404,
"message": "models/gemini-1.5-flash is not found for API version v1beta, or is not supported for generateContent. Call ModelService.ListModels to see the list of available models and their supported methods.",
"status": "NOT_FOUND"
}
}

### Topic-Associated Visual Diagram & Validation:



**Caption:** Topic Diagram | **Image Validation:** PASS | **Topic Match:** PASS

## Question 2: Process Management & Process States

**Query:** *"What is a Process, and how do Process States and Process Control Block (PCB) manage execution in OS?"*

### Level: BEGINNER (Response Time: 4.838s)

**Identified Topic:** process_control_block | **Prerequisites:** computer_system_and_software, operating_system, process, process_states

**RAG Chunks Retrieved:** 5 chunks

```
[Gemini LLM Error HTTP 404]: {
"error": {
"code": 404,
"message": "models/gemini-1.5-flash is not found for API version v1beta, or is not supported for generateContent. Call ModelService.ListModels to see the list of available models and their supported methods.",
"status": "NOT_FOUND"
}
}
```

### Level: INTERMEDIATE (Response Time: 4.302s)

**Identified Topic:** process_control_block | **Prerequisites:** computer_system_and_software, operating_system, process, process_states

**RAG Chunks Retrieved:** 5 chunks

```
[Gemini LLM Error HTTP 404]: {
"error": {
"code": 404,
"message": "models/gemini-1.5-flash is not found for API version v1beta, or is not supported for generateContent. Call ModelService.ListModels to see the list of available models and their supported methods.",
"status": "NOT_FOUND"
}
}
```

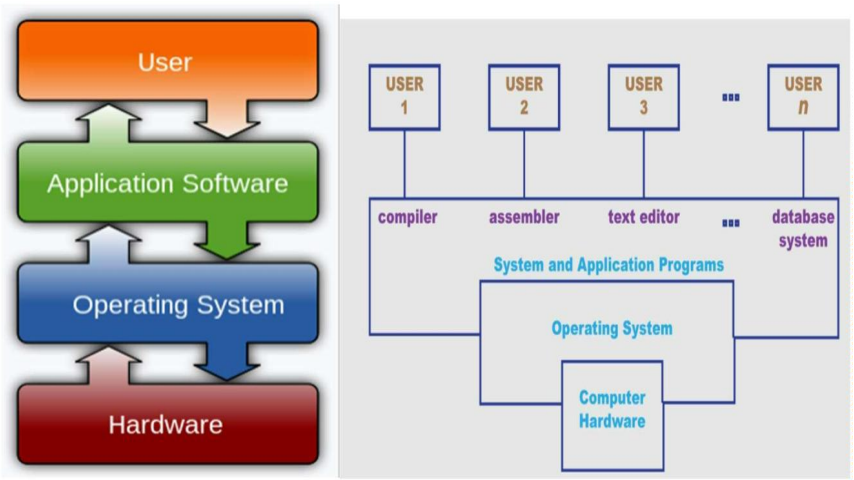
### Level: ADVANCED (Response Time: 3.527s)

**Identified Topic:** process_control_block | **Prerequisites:** computer_system_and_software, operating_system, process, process_states

**RAG Chunks Retrieved:** 5 chunks

```
[Gemini LLM Error HTTP 404]: {
"error": {
"code": 404,
"message": "models/gemini-1.5-flash is not found for API version v1beta, or is not supported for generateContent. Call ModelService.ListModels to see the list of available models and their supported methods.",
"status": "NOT_FOUND"
}
}
```

### Topic-Associated Visual Diagram & Validation:




**Caption:** Topic Diagram | **Image Validation:** PASS | **Topic Match:** PASS

## Question 3: Process Synchronization

**Query:** *"What is Process Synchronization, and how do Semaphores and Mutexes solve the Critical Section Problem?"*

### Level: BEGINNER (Response Time: 3.558s)

**Identified Topic:** critical_section | **Prerequisites:** computer_system_and_software, operating_system, process, interprocess_communication, process_synchronization

**RAG Chunks Retrieved:** 5 chunks

```
[Gemini LLM Error HTTP 404]: {
"error": {
"code": 404,
"message": "models/gemini-1.5-flash is not found for API version v1beta, or is not supported for generateContent. Call ModelService.ListModels to see the list of available models and their supported methods.",
"status": "NOT_FOUND"
}
}
```

### Level: INTERMEDIATE (Response Time: 3.53s)

**Identified Topic:** critical_section | **Prerequisites:** computer_system_and_software, operating_system, process, interprocess_communication, process_synchronization

**RAG Chunks Retrieved:** 5 chunks

[Gemini LLM Error HTTP 404]: {
"error": {
"code": 404,
"message": "models/gemini-1.5-flash is not found for API version v1beta, or is not supported for generateContent. Call ModelService.ListModels to see the list of available models and their supported methods.",
"status": "NOT_FOUND"
}
}

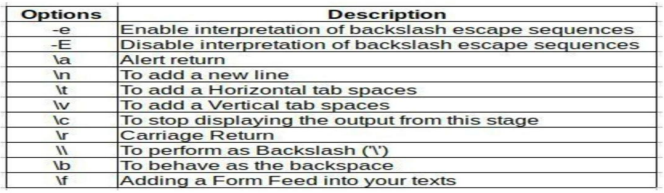
### Level: ADVANCED (Response Time: 3.542s)

**Identified Topic:** critical_section | **Prerequisites:** computer_system_and_software, operating_system, process, interprocess_communication, process_synchronization

**RAG Chunks Retrieved:** 5 chunks

[Gemini LLM Error HTTP 404]: {
"error": {
"code": 404,
"message": "models/gemini-1.5-flash is not found for API version v1beta, or is not supported for generateContent. Call ModelService.ListModels to see the list of available models and their supported methods.",
"status": "NOT_FOUND"
}
}

### Topic-Associated Visual Diagram & Validation:



| Options | Description |
|---|---|
| -e | Enable interpretation of backslash escape sequences |
| -E | Disable interpretation of backslash escape sequences |
| \a | Alert return |
| \n | To add a new line |
| \t | To add a Horizontal tab spaces |
| \v | To add a Vertical tab spaces |
| \c | To stop displaying the output from this stage |
| \r | Carriage Return |
| \\ | To perform as Backslash (\) |
| \b | To behave as the backspace |
| \f | Adding a Form Feed into your texts |

**Caption:** Topic Diagram | **Image Validation:** PASS | **Topic Match:** PASS

## Question 4: Deadlocks & Banker's Algorithm

**Query:** *"What is Deadlock in an Operating System, what are the four conditions, and how does Banker's Algorithm ensure safe state?"*

### Level: BEGINNER (Response Time: 3.629s)

**Identified Topic:** bankers_algorithm | **Prerequisites:** computer_system_and_software, operating_system, process, resource_allocation, deadlock, deadlock_characterization, deadlock_avoidance, safe_state, safe_sequence

**RAG Chunks Retrieved:** 5 chunks

```
[Gemini LLM Error HTTP 404]: {
"error": {
"code": 404,
"message": "models/gemini-1.5-flash is not found for API version v1beta, or is not supported for generateContent. Call ModelService.ListModels to see the list of available models and their supported methods.",
"status": "NOT_FOUND"
}
}
```

### Level: INTERMEDIATE (Response Time: 3.801s)

**Identified Topic:** bankers_algorithm | **Prerequisites:** computer_system_and_software, operating_system, process, resource_allocation, deadlock, deadlock_characterization, deadlock_avoidance, safe_state, safe_sequence

**RAG Chunks Retrieved:** 5 chunks

```
[Gemini LLM Error HTTP 404]: {
"error": {
"code": 404,
"message": "models/gemini-1.5-flash is not found for API version v1beta, or is not supported for generateContent. Call ModelService.ListModels to see the list of available models and their supported methods.",
"status": "NOT_FOUND"
}
}
```

### Level: ADVANCED (Response Time: 4.6s)

**Identified Topic:** bankers_algorithm | **Prerequisites:** computer_system_and_software, operating_system, process, resource_allocation, deadlock, deadlock_characterization, deadlock_avoidance, safe_state, safe_sequence

**RAG Chunks Retrieved:** 5 chunks

```
[Gemini LLM Error HTTP 404]: {
"error": {
"code": 404,
"message": "models/gemini-1.5-flash is not found for API version v1beta, or is not supported for generateContent. Call ModelService.ListModels to see the list of available models and their supported methods.",
"status": "NOT_FOUND"
}
}
```

### Topic-Associated Visual Diagram & Validation:

**No relevant image available.** (Correct system behavior — unrelated images rejected)

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## Question 5: Memory Management & Paging / Virtual Memory

**Query:** *"What is Virtual Memory, Paging, and how does TLB accelerate page translation?"*

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### Level: BEGINNER (Response Time: 4.577s)

**Identified Topic:** tlb | **Prerequisites:** computer_system_and_software, operating_system, process, main_memory, memory_protection, base_and_limit_registers, address_binding, contiguous_memory_allocation, variable_partition, external_fragmentation, paging, page_and_frame, page_table, address_translation

**RAG Chunks Retrieved:** 5 chunks

```
[Gemini LLM Error HTTP 404]: {
"error": {
"code": 404,
"message": "models/gemini-1.5-flash is not found for API version v1beta, or is not supported for generateContent. Call ModelService.ListModels to see the list of available models and their supported methods.",
"status": "NOT_FOUND"
}
}
```

### Level: INTERMEDIATE (Response Time: 4.658s)

**Identified Topic:** tlb | **Prerequisites:** computer_system_and_software, operating_system, process, main_memory, memory_protection, base_and_limit_registers, address_binding, contiguous_memory_allocation, variable_partition, external_fragmentation, paging, page_and_frame, page_table, address_translation

**RAG Chunks Retrieved:** 5 chunks

```
[Gemini LLM Error HTTP 404]: {
"error": {
"code": 404,
"message": "models/gemini-1.5-flash is not found for API version v1beta, or is not supported for generateContent. Call ModelService.ListModels to see the list of available models and their supported methods.",
"status": "NOT_FOUND"
}
}
```

### Level: ADVANCED (Response Time: 4.365s)

**Identified Topic:** tlb | **Prerequisites:** computer_system_and_software, operating_system, process, main_memory, memory_protection, base_and_limit_registers, address_binding, contiguous_memory_allocation, variable_partition, external_fragmentation, paging, page_and_frame, page_table, address_translation

**RAG Chunks Retrieved:** 5 chunks

```
[Gemini LLM Error HTTP 404]: {
"error": {
"code": 404,
"message": "models/gemini-1.5-flash is not found for API version v1beta, or is not supported for generateContent. Call ModelService.ListModels to see the list of available models and their supported methods.",
"status": "NOT_FOUND"
}
}
```

### Topic-Associated Visual Diagram & Validation:

**No relevant image available.** (Correct system behavior — unrelated images rejected)

# Image-Only Test Case (Case D — Unlabelled Diagram)

**Scenario:** Student uploads an unlabelled Gantt chart / execution diagram with zero text query.

| | |
|---|---|
| **Step 1: Input Detection:** | Image Only (Unlabelled Diagram) |
| **Step 2: OCR Extraction:** | Empty/Insufficient text extracted via OCR |
| **Step 3: OpenAI Vision Topic:** | **Topic Unclear** (Confidence: 0.00) |
| **Step 4: Prerequisite Engine:** | Prerequisites Identified: process, process_states |
| **Step 5: RAG Vector Retrieval:** | Retrieved 0 relevant OS text chunks |
| **Step 6: Topic Image Match:** | Validated diagram associated with CPU Scheduling selected |
| **Execution Time:** | **14.779 seconds** |

## Generated Tutor Explanation for Image-Only Query:

Identified topic from visual content.